INCH-POUND
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DETAIL SPECIFICATION

WEBBING AND TAPE, TEXTILE, ARAMID FIBER

This specification is approved for use by all Departments and Agencies of the Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers webbing and tape made from meta-aramid fiber.

1.2 Classification. This specification covers the following types as applicable.

Type I – Double woven webbing
Type II – Single woven tape

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Comments, suggestions, or questions on this document should be addressed: ATTN: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

AMSC N/A FSC 8305

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2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified. The issues are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies are available online at https://www.ftc.gov/)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AATCC Test Method 16.2 - Colorfastness to light: Carbon Arc
AATCC Test Method 16.3 - Colorfastness to Light: Xenon
AATCC Test Method 20 - Fiber Analysis: Qualitative
AATCC Test Method 61 - Colorfastness to Laundering: Accelerated
AATCC Test Method 81 - pH of Water-Extract from Wet Processed Textiles
AATCC Test Method 96 - Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool
AATCC Evaluation Procedure 1, Gray Scale for Color Change
AATCC Evaluation Procedure 2, Gray Scale for Staining
AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles

(Copies are available online at http://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 – Sampling Procedures and Tables for Inspection by Attributes

(Copies are available online at http://www.asq.org.)

ASTM INTERNATIONAL

ASTM D276 - Standard Test Method for Identification of Fibers in Textiles
ASTM D1776/D1776M - Standard Test Method for Conditioning and Testing Textiles
ASTM D1777 - Standard Test Method for Testing Thickness of Textile Materials

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ASTM D3775 - Standard Test Method for Fabric Count of Woven Fabric
ASTM D3776/D3776M - Standard Test Method for Mass Per Unit Area (Weight) of Fabric
ASTM F2203 - Standard Test Method for Linear Measurement Using Precision Steel Rule

(Copies of documents are available online at http://www.astm.org.)

PARACHUTE INDUSTRY ASSOCIATION (PIA)

PIA-TEST Method-4108 - Strength and Elongation, Breaking; Textile Webbing, Tape and Braided Items

(Copies of this document are available online at http://www.pia.com/)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. The finished webbing and tape shall match the standard sample for shade and appearance, and shall, unless otherwise indicated, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials.

3.4.1 Yarn. The yarn for the warp and filling shall be made from a bright continuous filament meta-aramid fiber.

3.5 Color. The color of the webbing and tape shall be Tan 499 or as specified in the contract or purchase order (see 6.2) when tested in accordance with 4.6.

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3.5.1 Colorfastness. The finished webbing and tape shall conform to the colorfastness requirements listed in Table I, unless otherwise stated in the contract or procurement documents when tested as specified in 4.6.

TABLE I. Colorfastness requirements (all Types).

Color Evaluation	Light (after 40 AFU or 170 kJ/(m^2nm)@420nm), (min.) 1/	Laundering (4 cycles) (min.)
Tan 499	3-4	3-4
All colors	3-4	3-4

1/ AFU: AATCC Fading Units

3.6 Physical requirements. The finished webbing and tape shall conform to the requirements, listed in Table II, when tested as specified in 4.6.

TABLE II. Physical requirements.

Characteristic	Type I	Type II
Width (inch)	5/8 (-1/32, +1/16)	1 (± 1/32)
Thickness (inch)	0.040 – 0.055	0.015 – 0.025
Weight, oz./lin.yd. (max)	0.45	0.30
Breaking strength, lb. (min.)	650	525
Ends in warp (min.)		
Face and back	158	276
Binder	9	--
Stuffer	40	--
Filling yarns per inch (min.)	48	30
Filling yarns per inch (min.) – shuttleless 1/	96	60

1/ Two picks per shed.

3.7 Weave. The weave for the webbing and tape shall be as specified in 3.7.1 and 3.7.2. When shuttleless construction is used a catchcord shall be utilized.

3.7.1 Type I. The webbing shall be a double fabric consisting of face and back warps with two ends weaving as one and a stuffer warp, all bound together by a binder warp with the ends weaving singly. The suffer ends shall weave between the face and back warps and shall not show on either the face or the back. The face warp shall weave plain weave with the picks that show on the face. The back warp shall weave plain weave with the picks that show on the back. The binder warp shall weave plain weave throughout.

3.7.2 Type II. The tape shall be plain weave, except that the warp ends shall weave two ends as one.

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3.8 Dimensional stability. The webbing and tape shall be heat set to meet the shrinkage of no more than 5.0 percent after five (5) cycles when tested as specified in 4.6.

3.9 pH. The pH value of the water extract of the finished tape or webbing shall not be less than 5.5 or more than 9.0 when tested as specified in 4.6.

3.10 Length and put-up. For Government procurement only, unless otherwise specified (see 6.2), the webbing and tape shall be furnished in lengths of 90 to 110-yards. No roll shall contain more than three (3) pieces and no piece shall be less than three (3) yards in length.

3.11 Fiber identification. Each roll of webbing and tape shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.12 Workmanship. The finished webbing and tape shall conform to the quality of product established by this specification. The quality acceptance level shall not exceed the established maximum value.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. A first article submitted in accordance with 3.1, shall be inspected, examined for appearance, color and finished defects and tested for the characteristics as specified in Table III and IV.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.5 and the tests of 4.6 through 4.7 as applicable. Sampling for inspection shall be performed in accordance with ANSI/ASQ Z1.4 and with quality acceptance limits as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2).

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4.5 Visual examination. The webbing and tape shall be examined for the defects listed in Table III.

TABLE III. Visual defects.

		Classification	
Examine	Defect	Major	Minor
Weave	Uneven weaving throughout		201
Twist or distortion	Webbing or tape will not lay flat upon application of manual pressure due to twist or distortion. 1/		202
Cut, hole, or tear	Any cut, hole, or tear.	101	
Spot, stain, or streak	Any clearly visible 2/		203
Drop-ply	Clearly visible on more than two ends within same length and extending over nine (9) linear inches or more. 2/	102	
	Clearly visible on one (1) or two (2) ends within same length and extending over nine (9) linear inches or more. 2/		204
Edges	Frayed, slack or otherwise poorly constructed and exceeding 1/4-inch in length.	103	
Knots	More than one (1) knot in any nine (9) linear inches.	104	
	More than one (1) knot every two (2)-yards with untrimmed ends extending from surface of webbing.		205
Mispick	Two or more across the full width.	105	
Uneven Width	Width not as specificed. Tight or loose filling resulting in variations of (±1/16) inch in width.		206 207
Color	Wrong color. Shade Bar 2/ Dye Streak 2/		208 209 210
Label	Mislabeled, not labeled in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.	106	
Length	More than three (3) pieces per roll Piece is less than three (3) yards in length		211 212

1/ A 3-yard length of webbing/tape shall be laid on a flat and smooth surface without tension. If the wedding tape does not lie flat, or if the wedding tape is still wavy or ridgy after applying manual pressure, it shall be counted as a defect.

2/ Clearly visible at normal inspection distance of three (3)-feet.

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4.6 End item testing. The webbing and tape shall be tested for the characteristics listed in Table I and II. The methods of testing as specified wherever applicable and as listed in Table IV shall be followed. All test reports shall contain the individual values utilized in expressing the final results. Unless otherwise specified, all testing shall be performed under standard conditions in accordance with ASTM D1776/D1776M. The lot size shall be expressed in units of 1 linear yard.

The sample unit for testing shall be 10 continuous yards of the finished webbing or tape for all physical and chemical tests. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. End item tests.

Characteristics	Requirement reference	Test method
Yarn identification	3.4.1	AATCC 20 or ASTM D276 1/ (see 6.4)
Visual shade matching	3.5	4.7.1
Colorfastness to:		
Light	Table I	AATCC 16.2, Option 1 or 16.3, Option 2, 2/
Laundering	Table I	AATCC 61, Test 1A, 3/
Width	Table II	ASTM F2203
Thickness	Table II	ASTM D1777, Option 2, 4/
Weight	Table II	ASTM D3776, Option D
Breaking strength	Table II	PIA Test Method 4108
Ends in warp:		
Face and back	Table II	ASTM D3775
Binder	Table II	ASTM D3775
Stuffer	Table II	ASTM D3775
Filling yarns per inch	Table II	ASTMD3775
Filling yarns per inch (shuttleless)	Table II	ASTM D3775
Weave	3.7	Visual
Dimensional stability (5 cycles)	3.8	AATCC 96 VIc, A 5/
pH	3.9	AATCC 81

1/ In case of dispute, the ASTM method prevails.
2/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change.
3/ Rated using the AATCC Evaluation Procedure 1, Gray scale for Color Change and AATCC Evaluation Procedure 2, Gray scale for Staining.
4/ The anvil shall be not less than 0.25-inches in diameter.
5/ Except that the length of the test specimen shall be 24-inches and the cotton laundering procedure shall be used. The stack temperature of the tumble drier shall be 180°F and the webbing or tape shall not be pressed.

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4.7 Methods of inspection.

4.7.1 Visual shade matching. The color and appearance of the webbing and tape shall match the standard sample when viewed using the ATTCC Evaluation Procedure 9, Option A, with sources simulating artificial daylight D75 illuminant with a color temperature of 7500 (± 200)K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856 (± 200)K.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory or contractually binding.)

6.1 Intended use. The Type I webbing is utilized as a slide fastener thong for the Shirt, Man's Army Aviation, Crewmember and as a waist adjustment strap for the trousers of Uniform, Army Aviation Crewmember. The Type II tape is utilized as a retrieval strap in Coveralls, Armored Vehicle, Crewmember.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this Specification
b. Type required (see 1.2)
c. The specific issue of individual documents referenced (see 2.2)
d. When first article is required (see 3.1 and 4.2)
e. Color required (see 3.5)
f. Length required if other than specified (see 3.10)
g. Conformance inspection acceptance quality limits (AQL) (see 4.3)
h. Inspection conditions (see 4.4)
i. Packaging (see 5.1)

6.3 Standard sample. For access to samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

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6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues, due to the extensiveness of the changes.

6.5 Subject term (key word) listing.

Meta
Uniforms
Yarn
Weave

Custodian:
Army – GL

Preparing Activity
DLA-CT

Review Activities:
Army – MD
Navy - NU

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